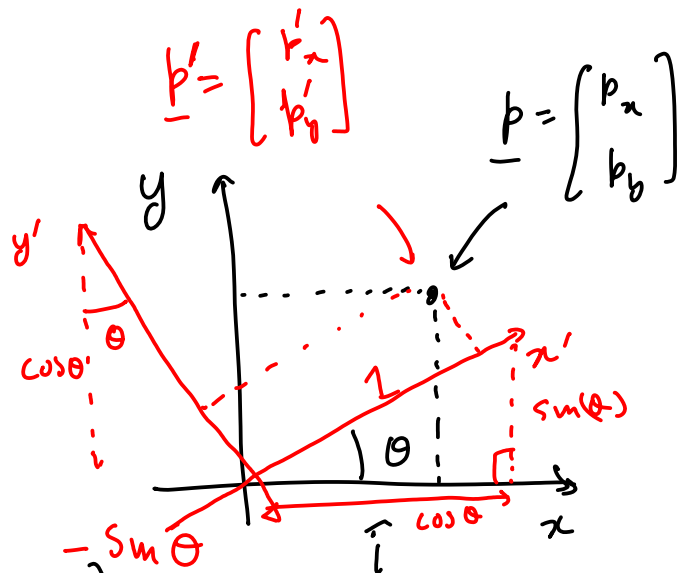


$$\underline{x}' = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \quad \underline{y}' = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$



$$\begin{aligned} \underline{p} &= p_x \underline{x} + p_y \underline{y} = \begin{bmatrix} \underline{x} & \underline{y} \end{bmatrix} \begin{bmatrix} p_x \\ p_y \end{bmatrix} \\ &= p'_x \underline{x}' + p'_y \underline{y}' = \begin{bmatrix} \underline{x}' & \underline{y}' \end{bmatrix} \begin{bmatrix} p'_x \\ p'_y \end{bmatrix} \end{aligned}$$

$$\underline{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \underline{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \underline{x} & \underline{y} \end{bmatrix} \underline{p} = \begin{bmatrix} \underline{x}' & \underline{y}' \end{bmatrix} \begin{bmatrix} p'_x \\ p'_y \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \underline{p} = \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}}_{R(\theta)} \underline{p}'$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} c^2 \theta + s^2 \theta & -(\cancel{c} s \theta + s \cancel{c} \theta) \\ \dots & c^2 \theta + s^2 \theta \end{bmatrix}$$

$$\begin{bmatrix} c \theta & s \theta \\ -s \theta & c \theta \end{bmatrix} \begin{bmatrix} c \theta & -s \theta \\ s \theta & c \theta \end{bmatrix}$$

$$\underline{p} = R(\theta) \underline{p}'$$

$$\underbrace{R^T R = \underline{I}}_{\text{orthonormal matrix}} \Rightarrow R^{-1} = R^T$$

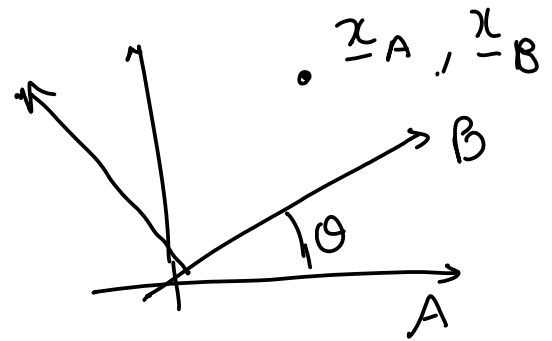
$$\det(R) = 1$$

$$\underline{p}' = R^T(\theta) \underline{p}$$

Coordinates (coordinate frames)

$$\underline{x}_B = {}^B R_A \underline{x}_A$$

$$\underline{x}_A = {}^A R_B \underline{x}_B$$

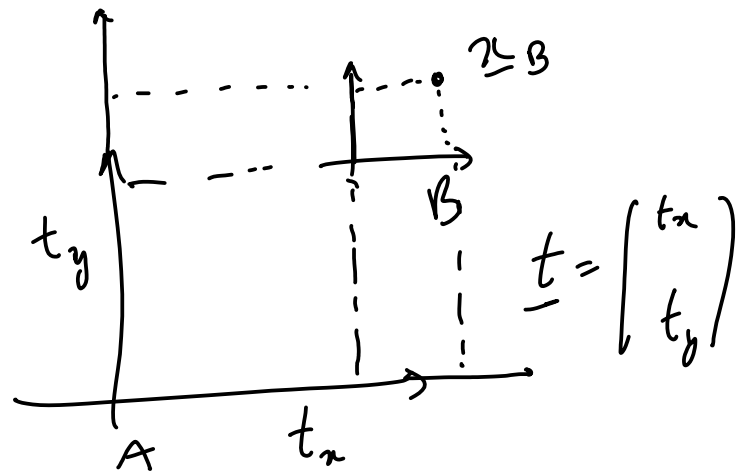


Target $\rightarrow$ ${}^A R_B = R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$\uparrow$
source

Translation

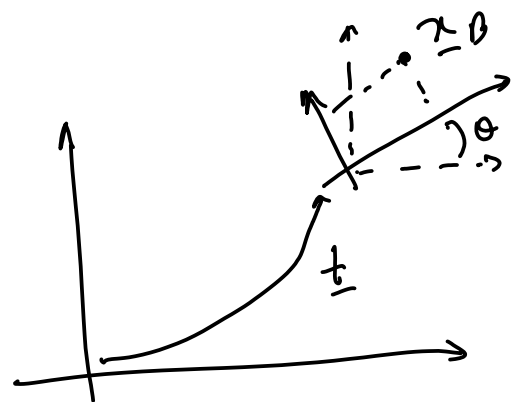
$$\underline{x}_A = \underline{t} + \underline{x}_B$$



$$\underline{x}_A = R(\theta) \underline{x}_B + \underline{t}$$

$$\underline{x}_A = {}^A R_B \underline{x}_B + {}^A \underline{t}_B$$

Transformation matrix



$$\begin{bmatrix} \underline{x}_A \\ 1 \end{bmatrix}_{3 \times 1} = \underbrace{\begin{bmatrix} {}^A R_B & \underline{t} \\ 0^T & 1 \end{bmatrix}}_{A \rightarrow B} \begin{bmatrix} \underline{x}_B \\ 1 \end{bmatrix}_{3 \times 1}$$

$\begin{matrix} 2 \times 2 & 2 \times 1 \\ 1 \times 2 & 1 \end{matrix}$

homogenized coordinate

$$\tilde{\underline{x}}_A = {}^A T_B \tilde{\underline{x}}_B$$

$$\underline{x}_A = \begin{bmatrix} {}^A R_B & \underline{t} \\ \hline 1 \end{bmatrix} \underbrace{\begin{bmatrix} \underline{x}_B \\ 1 \end{bmatrix}}_{\tilde{\underline{x}}_B}$$

✓